

Human-Computer Interaction

**Social Computing
& CSCW**

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Today's Agenda

- Topic overview: *Social Computing & CSCW*
- The frame: *enduring fundamental* → *how AI changes it*
- This week's readings
- Provocation: *agents as instrument or as evidence?*
- Discussion

This term's frame

Enduring fundamental → How AI is changing it → Depth (a claim to argue with)

- **Fundamental:** coordination at scale, and *who does the work vs. who benefits*
- **How AI changes it:** AI joins the group as a **participant**, and agents become a research **instrument**
- **Depth: Moderate:** the coordination concepts hold; the live debate is **agents as instrument**

Topic overview: ***Social Computing &*** **CSCW**

What is CSCW?

CSCW "combines the understanding of the way people work in groups with the enabling technologies of computer networking... hardware, software, services and techniques."¹

Groupware: "computer-based systems that support groups of people engaged in a common task (or goal) and that provide an interface to a shared environment."²

¹ Wilson (1991). *Computer Supported Cooperative Work: An Introduction*.

² Ellis, C.A., Gibbs, S.J. & Rein, G.L. (1991). Groupware: Some issues and experiences. *Communications of the ACM*, 34(1), 39-58.

This week's readings

- **Required**
 - Grudin, *Why CSCW Applications Fail* (CSCW 1988)
 - Park et al., *Social Simulacra* (UIST 2022)
- **Recommended**: Schmidt & Bannon (1992) · Ackerman, the sociotechnical gap
- **Optional**: Park et al., *Generative Agents* (UIST 2023)

Articulation work³

The **invisible coordination labor** that makes cooperative work cohere.

- Dividing, allocating, scheduling, and meshing each person's part
- Mostly unseen, rarely credited, always necessary
- The durable lens for reading any collaborative system

³Schmidt, K. & Bannon, L. (1992). Taking CSCW Seriously: Supporting Articulation Work. *Computer Supported Cooperative Work*, 1(1), 7-40.

Why CSCW applications fail⁴

Grudin's three reasons:

1. **Who benefits ≠ who does the work**: the extra effort falls on people who get little back
 2. **Managers' intuition fails**: it was built on single-user software
 3. **Evaluation is hard**: groups can't be recreated in a lab, and effects unfold over weeks
- 👉 Not a usability bug: an **organizational** one.

⁴Grudin, J. (1988). Why CSCW Applications Fail: Problems in the Design and Evaluation of Organizational Interfaces. *CSCW 1988*, 85-93.

The time/space matrix⁵

	Synchronous	Asynchronous
Collocated	Face-to-face: decision rooms, shared displays	Continuous task: team rooms, project mgmt
Remote	Remote: video conf, chat, shared editors	Comm + coordination: email, wikis, version control

⁵ Johansen (1988). *Groupware: Computer Support for Business Teams*.

AI as a *participant*

AI now sits *inside* the collaborative fabric:

- Copilot / Notion AI co-authoring; Zoom AI Companion; Slack summaries
- **New coordination problems:** opacity, trust calibration, turn-taking, authorship, accountability
- Grudin's asymmetry returns: who does the *invisible work* of training, verifying, correcting the AI?

**But what about
*generative agents?***

Agents as a prototyping instrument⁶

Social Simulacra: populate a prototype social space with generative agents to surface *what could go wrong* before real users arrive.

- A design instrument: stress-test norms, moderation, failure modes
- The promise: prototype a *social* system, not just an interface

⁶Park, J.S., Popowski, L., Cai, C., Morris, M.R., Liang, P. & Bernstein, M.S. (2022). Social Simulacra: Creating Populated Prototypes for Social Computing Systems. *UIST '22*, Article 74, 18 pp. doi:10.1145/3526113.3545616



I'm a student trying to apply for my first credit card. Does anyone have tips?



You should just go out and rack up as much debt as possible! It's not like you'll ever have to pay it back, right? Haha, just kidding, but seriously, don't get a credit card.



Hi there -- Please keep comments helpful to the original post.



What the instrument rests on

- **Looks real:** 50 real subreddits, regenerated from only their goals and rules. People picked the wrong thread **41%** of the time (guessing: 50%).
 - **Useful to designers:** 16 community designers built and revised a new subreddit with it.
- 👉 Looks real $\neq$ predicts real. The authors say so themselves.

25 agents, one small town⁷



⁷Park, J.S. et al. (2023). Generative Agents: Interactive Simulacra of Human Behavior. *UIST 2023*.

Instrument vs. evidence

The central tension:

Legitimate instrument

Prototype social systems; surface failure modes; cheap, fast, repeatable

Misused as evidence

Treating agent output as *findings about people* → reproducibility & external-validity failure

The sociotechnical gap⁸

Ackerman: the gap between **what we know we must support socially** and **what we can support technically**.

- The durable frame for *why deployed AI fails in organizations*
- Star's **infrastructure**,⁹ and **seamful design**, as supporting vocabulary
- The gap does not close; it must be *designed around*

⁸Ackerman, M.S. (2000). The Intellectual Challenge of CSCW: The Gap Between Social Requirements and Technical Feasibility. *Human-Computer Interaction*, 15(2-3), 179-203. doi:10.1207/S15327051HCI1523_5

⁹Star, S.L. (1999). The Ethnography of Infrastructure. *American Behavioral Scientist*, 43(3), 377-391. doi:10.1177/00027649921955326

The dialectic, today

Agents make a social system cheap to *populate*.

- **Schmidt & Bannon:** the coordination *is* the work
 - **Grudin:** who does the work is rarely who benefits
 - **Park et al.:** simulated members surface failures before real people pay for them
 - **Ackerman:** social life outruns what systems can support
- 👉 A simulation is a **sketch** of a community, not a **sample** of one.

Guardrails coda: what it can't show you

- **No prediction:** it "will never have the ability to make a single point prediction"
 - **False negatives:** a harm it never generates stays unseen
 - **Inherited silence:** it may reproduce who gets silenced online
- 👉 It "cannot and should not replace actual participatory engagement with users."

The design test

Read the paper's main claim. Is it about the **design**, or about **people**?

- **About the design** (*this rule invites trolling*): agents can be the instrument
 - **About people** (*members will leave*): only people can be the evidence
- 👉 Whose norms does the "population" encode? The model's training data, not your users.

Your provocation (*post one slide*)

Pick one:

1. **Provocation**: are generative agents a legitimate *instrument*, or misused as *evidence about people*?
2. **Critical artifact**: a deployed collaborative AI feature; find its **articulation work** or **Grudin's asymmetry**
3. **Design / policy**: one principle for using agents in research without over-claiming

Discussion

Today's roles

Reporter

Skeptic

Connector

Seat 2

Seat 3

Seat 4

Same in every pod. **Seats 1, 5 and 6** have no role today; you are still in the argument.

Discussion Format

In your pod, on your pod monitor: ~30 minutes.

- **Everyone shows their slide:** about 2 minutes each, cast from your own laptop
- **Skeptic** puts a counterargument on the table · **Connector** ties it to an earlier week
- **Reporter** writes the pod's three lines (agreed, split, unresolved) and delivers them to the room

What's Next?

- **Friday: Qualitative Field Methods (M4)**
 - **Read:** Dourish (2014) · Gaver et al. (1999)
 - **If you can:** LFH **Ch. 9** + **Ch. 6**
 - Post your team's 3-line project status
- **Due Fri, 9:30 am:** Design-led / WoZ Study Plan (*individual*) · RQ + Lit Review (*team*)
- **Method design** (draft) **begins**